

Program 10 : Demonstrate Kubernetes Dashboard and CLI for ClusterMonitoring

Manifet Files

```
abhi@abhi-ThinkPad-T490:~/devops/program10$ cat Dep.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
abhi@abhi-ThinkPad-T490:~/devops/program10$ cat Svc.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: NodePort
  selector:
    app: nginx
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30080
abhi@abhi-ThinkPad-T490:~/devops/program10$ k apply -f Dep.yaml
deployment.apps/nginx-deployment created
abhi@abhi-ThinkPad-T490:~/devops/program10$ k apply -f Svc.yaml
service/nginx-service created
abhi@abhi-ThinkPad-T490:~/devops/program10$ k get pods
NAME                                READY   STATUS              RESTARTS   AGE
my-cronjob-29435295-z64s2           0/1     Completed           0           7m52s
my-cronjob-29435300-qm7jz           0/1     Completed           0           4m32s
nginx-deployment-6f9664446b-pcmzp  0/1     ContainerCreating   0           7s
nginx-deployment-6f9664446b-s6qk8  0/1     ContainerCreating   0           7s
abhi@abhi-ThinkPad-T490:~/devops/program10$ k get deployments
NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
```

Deployments and Pod Description

Welcome to nginx!

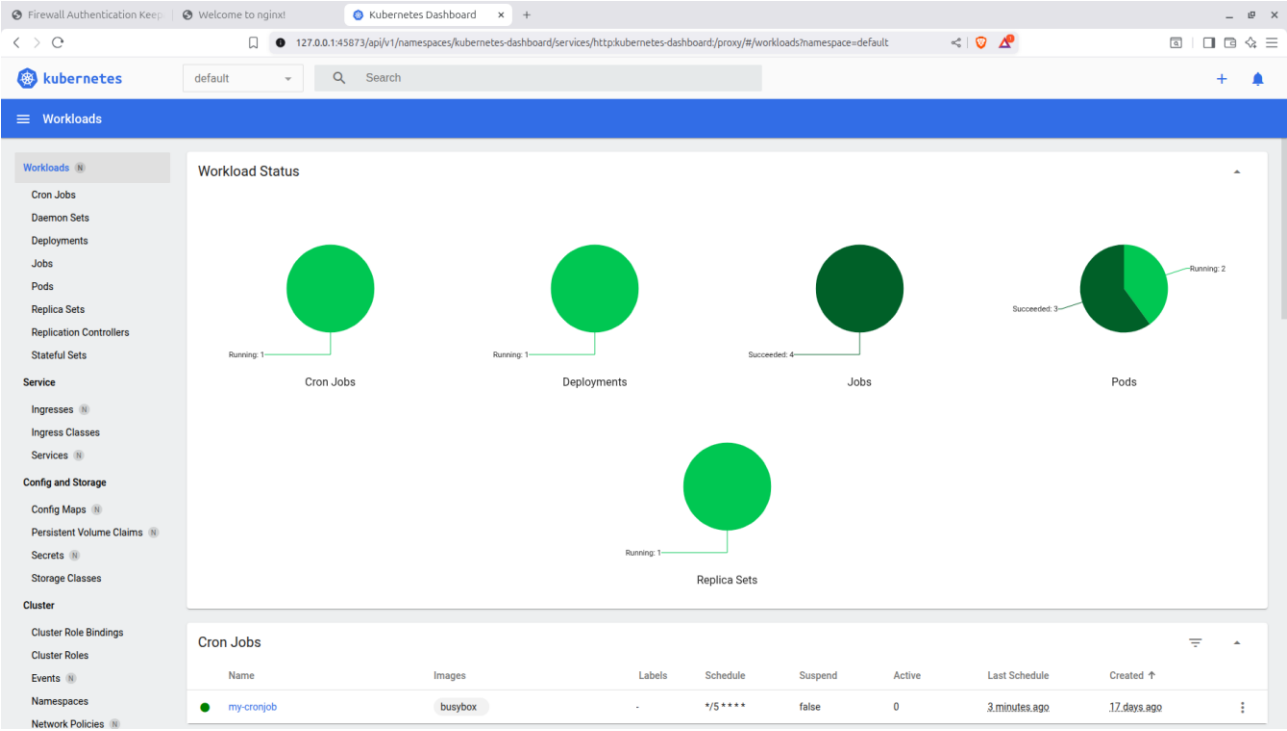
If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

```
nginx-deployment-6f9664446b-s6qk8 0/2 ContainerCreating
abhi@abhi-ThinkPad-T490:~/devops/program10$ k get deployments
NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment                    0/2     2             0           11s
abhi@abhi-ThinkPad-T490:~/devops/program10$ k describe pod nginx-deployment-6f9664446b-s6qk8
Name:                               nginx-deployment-6f9664446b-s6qk8
Namespace:                           default
Priority:                             0
Service Account:                       default
Node:                                 minikube/192.168.49.2
Start Time:                           Fri, 19 Dec 2025 09:54:25 +0530
Labels:                               app=nginx
                                      pod-template-hash=6f9664446b
Annotations:                           <none>
Status:                               Running
IP:                                   10.244.0.75
IPs:
  IP:                                 10.244.0.75
Controlled By:                         ReplicaSet/nginx-deployment-6f9664446b
Containers:
  nginx:
    Container ID:   docker://2a29d598efcc393fffd4881f3fc6802eab4a24f0e9012f076a728b2615966704d
    Image:          nginx:latest
    Image ID:       docker-pullable://nginx@sha256:fb01117203ff38c2f9af91db1a7409459182a37c87cced5cb442d
    Port:          80/TCP
    Host Port:     0/TCP
    State:         Running
      Started:     Fri, 19 Dec 2025 09:54:35 +0530
    Ready:         True
    Restart Count: 0
    Environment:   <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-grpct (ro)
Conditions:
  Type                               Status
  PodReadyToStartContainers         True
  Initialized                       True
  Ready                             True
  ContainersReady                   True
  PodScheduled                      True
Volumes:
  kube-api-access-grpct:
    Type:              Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName:       kube-root-ca.crt
    Optional:            false
```

Kubernetes Dashboard



Name	Images	Labels	Schedule	Suspend	Active	Last Schedule	Created
my-cronjob	busybox	-	*/* * * * *	false	0	3 minutes ago	17 days ago